Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

Team ID :

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Project Final Report

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1. <u>INTRODUCTION</u>

1.1 Project Overview :

In today's competitive food delivery industry, understanding customer ordering behaviour has become essential for restaurants, cloud kitchens, and food delivery platforms to improve business performance and customer satisfaction. Every customer order generates valuable information such as order value, cuisine preference, delivery time, customer ratings, demographics, and purchasing patterns. Converting this large volume of raw transactional data into meaningful insights enables businesses to make informed decisions rather than relying on assumptions.

This project, **"Food Ordering Behaviour and Consumer Trends: An Interactive Tableau Dashboard for Restaurant Business Intelligence,"** focuses on analysing food ordering data to uncover valuable business insights. The project processes a large dataset containing approximately **50,000 customer transactions**, performs data preparation and transformation, and presents the results through an interactive business intelligence dashboard developed using **Tableau Public**. The dashboard is seamlessly integrated into a **Flask-based web application**, providing users with a responsive and user-friendly interface for exploring key business metrics.

The application presents important Key Performance Indicators (KPIs) such as Total Orders, Total Revenue, Average Order Value, Average Delivery Time, Customer Ratings, and Repeat Order Percentage. Interactive filters, dynamic visualizations, and dashboard components allow users to analyse customer behaviour across different cities, cuisines, meal types, restaurant categories, weather conditions, and customer demographics. The integrated Tableau dashboards enable stakeholders to identify high-performing locations, monitor delivery efficiency, evaluate customer satisfaction, and recognize purchasing trends that support operational and strategic decision-making.

By combining Python Flask for web deployment with Tableau Public for interactive data visualization, the project delivers a modern business intelligence solution that transforms complex food ordering data into clear, actionable insights.
The platform assists restaurant owners, business analysts, marketing teams, and decision-makers in improving operational efficiency, enhancing customer experience, and developing data-driven business strategies for sustained growth.

1.2 Purpose :

The purpose of the **"Food Ordering Behaviour and Consumer Trends: An Interactive Tableau Dashboard for Restaurant Business Intelligence"** project is to transform large volumes of food ordering data into meaningful business insights through an interactive web-based analytics platform. By integrating **Python Flask** with **Tableau Public**, the project provides a centralized environment where restaurant owners, business analysts, and decision-makers can monitor key performance indicators (KPIs), explore customer behaviour, and evaluate business performance efficiently.

The project aims to:

- Help restaurant owners identify the cities, cuisines, meal types, and restaurant categories that generate the highest number of orders and revenue.
- Enable business analysts to monitor customer ordering patterns, spending behaviour, and repeat purchase trends using interactive dashboards.
- Assist delivery management teams in analysing delivery performance and identifying opportunities to improve service efficiency and customer satisfaction.
- Support marketing teams in understanding customer demographics, seasonal trends, and behavioural patterns to develop targeted promotional campaigns.
- Analyse the influence of order value, customer ratings, delivery time, discounts, weather conditions, and restaurant type on overall business performance.
- Provide dynamic filtering and interactive visualizations that allow stakeholders to compare business metrics across multiple dimensions and make informed decisions.

- Deliver an intuitive web-based dashboard that presents real-time business intelligence through embedded Tableau visualizations without requiring users to interact directly with raw datasets.
- Promote data-driven decision-making by replacing manual spreadsheet analysis with an integrated business intelligence solution.

In summary, the project provides an efficient and user-friendly platform for analysing food ordering behaviour, monitoring operational performance, and supporting strategic business decisions. The combination of Flask-based web deployment and Tableau-powered interactive dashboards enables stakeholders to transform raw transactional data into actionable insights that improve customer experience, operational efficiency, and overall business growth.

2. IDEATION PHASE

2.1 Problem Statement :

A well-defined problem statement ensures that the project remains focused on solving real business challenges faced by its intended users. In this project, the primary users include restaurant owners, business analysts, delivery managers, and marketing teams who need a simple and efficient way to analyse food ordering behaviour and business performance. Although large volumes of transactional data are generated every day, extracting meaningful insights from raw datasets is difficult due to scattered information, inconsistent data, and the absence of interactive analytical tools.

The project analyses food ordering data containing customer demographics, cuisine preferences, restaurant categories, order values, delivery performance, ratings, discounts, weather conditions, and repeat purchase behaviour. By integrating an interactive Tableau dashboard within a Flask web application, users can visualize key performance indicators (KPIs), compare trends across multiple business dimensions, and make informed decisions related to operations, customer engagement, pricing strategies, and business growth.

Customer Problem Statement Table

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	**A Business Analyst**	Analyse food ordering behaviour, customer preferences, and business performance using interactive dashboards.	It is difficult to identify meaningful trends and KPIs from large volumes of food ordering data.	The raw dataset is complex, contains multiple variables, and lacks clear visual representation for decision-making.	Frustrated
PS-2	**An Operations Manager**	Improve delivery performance, monitor operational efficiency, and reduce delivery delays.	It is challenging to identify delivery bottlenecks, peak order periods, and service performance issues.	The available operational data is scattered and not presented through interactive dashboards with dynamic filtering.	Concerned
PS-3	**A Restaurant Owner**	Increase revenue, identify popular cuisines, and improve customer satisfaction.	It is difficult to determine which cities, cuisines, and restaurant categories contribute the most to business growth.	Traditional reports do not provide consolidated KPIs or business intelligence insights for strategic planning.	Uncertain

Customer Problem Summary

Restaurant owners and business stakeholders require an easy-to-use, web-based business intelligence solution that transforms complex food ordering data into interactive visualizations and meaningful insights.

The platform should enable users to monitor KPIs, identify customer trends, evaluate operational performance, and support data-driven decision-making without requiring advanced technical knowledge.

Impact

- Poor business decisions due to the lack of centralized analytical dashboards.
- Difficulty identifying high-performing cities, cuisines, restaurant categories, and customer segments.
- Time-consuming manual analysis of large datasets using spreadsheets.
- Reduced operational efficiency caused by limited visibility into delivery performance and customer satisfaction.
- Missed opportunities for targeted marketing campaigns, pricing optimization, and revenue growth.
- Inability to quickly monitor business KPIs and customer behaviour through a single interactive platform.

2.2 Empathy Map Canvas :

An empathy map provides a structured understanding of the users' behaviours, needs, thoughts, and challenges while interacting with business data. For this project, it helps restaurant owners, business analysts, operations managers, and marketing teams understand food ordering trends, customer preferences, delivery performance, and revenue patterns through an interactive Tableau dashboard integrated with a Flask web application.

Before developing the dashboard, it is important to understand the perspective of the end users—their goals, frustrations, daily activities, and expectations. This understanding ensures that the dashboard addresses real business problems and delivers meaningful insights that support data-driven decision-making.

Develop Shared Understanding and Empathy

THINK & FEEL	HEAR	SEE	SAY & DO
Which cuisines are most popular?	Customers want faster delivery and better service.	High demand for certain cuisines and meal types.	I need to analyze order trends regularly.
How can we increase revenue?	Food quality and taste matter a lot.	Peak order times in weekends and evenings.	I monitor performance using dashboards and reports.
Are customers satisfied with delivery time?	Discounts and offers influence ordering decisions.	Delivery time variations across cities.	I compare cities, cuisines and meal types.
Which cities contribute the most orders?	People prefer more variety in cuisines.	Ratings and reviews affect customer choices.	I track customer ratings and delivery time.
How can we improve customer experience?	Customers compare prices across platforms.	Revenue contribution by meal types and cuisines.	I use insights to make data-driven business decisions.
What factors affect repeat orders?		Repeat orders from satisfied customers.	I share reports with the team for better planning.

PAIN	GAIN
Difficult to analyze large amounts of raw data.	Interactive dashboards provide clear insights.
Time-consuming manual reporting process.	Quickly identify popular cuisines and high-demand cities.
Hard to identify customer preferences quickly.	Monitor delivery performance and customer satisfaction.

PAIN	GAIN
Delivery delays affect customer satisfaction and ratings.	Data-driven decisions improve revenue and efficiency.
Lack of real-time insights for quick decisions.	Better customer experience leads to more repeat orders.

2.3 Brainstorming :

Step-1: Team Gathering, Collaboration and Select the Problem Statement

- **Problem Statement**: "How can food ordering data be analyzed and visualized to identify customer food habits, cuisine preferences, delivery performance, and revenue across different cities?"

Step-2: Brainstorm, Idea Listing and Grouping

The brainstorming session resulted in several ideas categorized into the following groups:

- **Customer Insights**: Including customer demographics, age, and mood analysis.
- **Business Performance**: Focusing on revenue trends, average order value, and performance metrics.
- **Logistics & Delivery**: Analyzing delivery times and efficiency.
- **Visualization & Dashboard**: Planning the specific visual output, such as KPI cards, heatmaps, and trend charts.
- **Project Plan**: Identifying the necessary research and documentation steps.

Step-3: Idea Prioritization

The team utilized a prioritization matrix based on **Feasibility** and **Impact**:

- **Do First (High Impact, High Feasibility)**:
    - KPI cards (Orders, Revenue, Average Order Value).
    - Trend analysis charts and heatmaps.
- **Plan Next (High Impact, Lower Feasibility)**:
    - Developing comprehensive dashboard visualizations.
    - Detailed business performance reporting.

- **Do Later (Lower Impact, High Feasibility)**:
    - Logistics and delivery time analysis.
    - Customer demographic breakdown.
- **Maybe (Lower Impact, Lower Feasibility)**:
    - Predictive modeling or advanced analytics.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map :

The Customer Journey Map illustrates the complete experience of a business analyst, restaurant owner, or operations manager while using the Food Ordering Behaviour and Consumer Trends Dashboard. It describes how users interact with the Flask-based web application and Tableau dashboards to transform raw food ordering data into meaningful business insights. The journey begins when users recognize the need for better business intelligence, continues through interactive dashboard exploration, and concludes with data-driven decision-making that improves operational efficiency, customer satisfaction, and business performance. Understanding this journey helps ensure that the dashboard addresses real user needs and provides a seamless analytical experience.

Stage	Steps	Interactions	Goals & Motivations
Entice	Business analyst identifies the need to understand customer ordering behaviour and business performance.	Business reports, Tableau dashboard link, project portal.	Identify business opportunities, customer trends, and revenue patterns.
Enter	User opens the Flask web application and accesses the interactive Tableau dashboard.	Web browser, Flask application, embedded Tableau dashboard.	Access the analytics platform quickly and securely.
Engage	User applies filters to analyse orders by city, cuisine, meal	Interactive dashboard, filters,	Discover customer preferences, monitor

Stage	Steps	Interactions	Goals & Motivations
	type, customer age, ratings, and delivery performance.	KPI cards, charts, and tooltips.	KPIs, and identify business insights.
Exit	User reviews dashboard findings, generates reports, and shares insights with stakeholders.	Tableau Story, downloadable reports, dashboard screenshots.	Present analytical findings clearly for business decision-making.
Extend	Restaurant management implements recommendations and continuously monitors business performance using updated dashboards.	Business review meetings, dashboard refresh, periodic analysis.	Improve operational efficiency, customer satisfaction, and revenue growth through data-driven decisions.

3.2 Solution Requirement :

The **Food Ordering Behaviour and Consumer Trends** dashboard is designed to provide an interactive business intelligence platform that enables restaurant owners, business analysts, and management teams to analyse customer ordering behaviour and operational performance effectively. The solution combines a **Flask-based web application** with **Tableau Public dashboards** to deliver meaningful insights through interactive visualizations, KPI cards, filters, and reports. To ensure that the application meets both business and technical objectives, the project requirements are categorized into **Functional Requirements**, which describe the core features and services provided by the system, and **Non-Functional Requirements**, which define the quality attributes such as usability, reliability, performance, security, and scalability. These requirements serve as the foundation for designing, developing, testing, and deploying a robust business intelligence solution that supports data-driven decision-making.

Functional Requirements :

FR No.	Functional Requirement (Epic)	Sub-Requirement (Story / Sub-Task)
FR-1	**Dashboard Home & Navigation**	1. Display the project home page using Flask.2. Provide navigation between dashboard sections.3. Ensure responsive and user-friendly interface.
FR-2	**Interactive Dashboard Integration**	1. Embed Tableau dashboard into the Flask web application.2. Display KPI cards and interactive charts.3. Enable dashboard rendering using Tableau JavaScript API.
FR-3	**Customer Behaviour Analysis**	1. Analyse food ordering behaviour by city, cuisine, meal type, and customer demographics.2. Compare ordering patterns across different customer segments.3. Identify popular cuisines and high-demand locations.
FR-4	**Business Performance Analysis**	1. Visualize Total Orders, Revenue, Average Order Value, Delivery Time, and Customer Ratings.2. Compare business performance across restaurant categories.3. Monitor delivery efficiency and repeat customer behaviour.
FR-5	**Interactive Filtering & Reporting**	1. Enable filters for city, cuisine, meal type, restaurant type, and customer age.2. Provide drill-down analysis using Tableau dashboards.3. Present business insights through dashboards and Tableau Story.

Non-Functional Requirements :

NFR No.	Requirement	Description
NFR-1	**Usability**	Provides an intuitive Flask-based interface with interactive Tableau dashboards that require minimal user training.
NFR-2	**Security**	Ensures secure management of application files, Flask configuration, and dashboard resources with controlled access.

NFR No.	Requirement	Description
NFR-3	Reliability	Displays consistent and accurate analytical results through validated and cleaned datasets.
NFR-4	Performance	Loads dashboard pages quickly and supports smooth interaction with filters, charts, and KPI visualizations.
NFR-5	Availability	Makes the analytics dashboard accessible through a web browser and Tableau Public whenever required.
NFR-6	Scalability	Supports additional datasets, dashboard components, and analytical features without major application redesign.

3.3 Data Flow Diagram :

The Food Ordering Behaviour and Consumer Trends system follows a structured data flow model to transform raw food ordering data into meaningful business intelligence through an interactive Flask web application integrated with Tableau dashboards. The system processes customer transaction data, performs data preparation, generates analytical visualizations, and presents business insights to decision-makers through an easy-to-use web interface.

- **DFD Level 0 (Context Diagram)**: This high-level diagram illustrates the interaction between the Food Ordering Behaviour and Consumer Trends Dashboard System and its primary users, including the Restaurant Owner, Business Analyst, Operations Manager, and Marketing Manager. The system receives the Food Ordering Dataset (50,000 Records) as input and provides interactive dashboards, KPI reports, and business insights as output.
- **DFD Level 1 (Process Breakdown Flow)**: This diagram describes the internal workflow of the application:
  1. 1.0 Data Preparation: Cleans the dataset by handling missing values, duplicate records, and inconsistent data.
  2. 2.0 Data Processing & Analysis: Performs data transformation, aggregation, KPI calculation, and customer behaviour analysis.
  3. 3.0 Tableau Dashboard Integration: Creates interactive dashboards, charts, filters, and Tableau Story visualizations.

4. 4.0 Flask Web Deployment: Embeds the Tableau dashboard within the Flask application and delivers analytical reports to business users through the web interface.

3.3 Technology Stack :

Technical Architecture

The **Food Ordering Behaviour and Consumer Trends** system follows a three-layer architecture that integrates data visualization with web application technologies to provide an interactive business intelligence platform.

- **User Layer:** Consists of restaurant owners, business analysts, operations managers, and marketing teams who access the dashboard through a modern **web browser**. The responsive user interface enables users to explore business insights using interactive filters, KPI cards, charts, and dashboards.
- **Application & Analytics Layer:** The core application is developed using the **Flask framework**, which manages routing, webpage rendering, and dashboard integration. Interactive dashboards are created using **Tableau Desktop** and published through **Tableau Public**, enabling users to visualize food ordering trends and business performance.
- **Data & Storage Layer:** The system processes the **Food Ordering Behaviour Dataset** stored in CSV format. Data cleaning, transformation, and visualization are performed before embedding the dashboard into the Flask application. Project source code and datasets are maintained using **GitHub** for version control and collaboration.

Table-1: Components & Technologies :

S.No	Component	Description	Technology
1	User Interface	Responsive web interface for dashboard access	HTML5, CSS3, Bootstrap 5
2	Backend Framework	Web application development and routing	Python Flask
3	Programming Language	Backend application logic	Python 3.x
4	Data Source	Food Ordering Behaviour dataset	CSV File
5	Data Preparation	Data cleaning, preprocessing and transformation	Tableau Desktop
6	Data Visualization	Interactive charts, KPIs and graphs	Tableau Desktop
7	Dashboard	Interactive business intelligence dashboard	Tableau Dashboard
8	Story Visualization	Presentation of analytical insights	Tableau Story
9	Database / Storage	Structured transactional dataset	CSV
10	Version Control	Source code and project management	GitHub
11	Cloud Platform	Online dashboard publishing	Tableau Public
12	Development Environment	Application development and testing	Visual Studio Code
13	Dashboard Integration	Embedding Tableau dashboards into Flask	tableau-2.min.js (Tableau JavaScript API)

Table-2: Application Characteristics :

S.No	Characteristics	Description	Technology
1	Open-Source Framework	Uses open-source technologies for web application development	Python, Flask, HTML, CSS, Bootstrap
2	Security	Secure handling of application files and dashboard resources	Flask Configuration, HTTPS
3	Interactive Analytics	Supports dynamic filtering and interactive dashboard exploration	Tableau Dashboard
4	Scalability	Supports larger datasets and additional analytical modules	Flask Architecture, Tableau BI Engine
5	Availability	Dashboard accessible through any modern web browser	Flask Web Server, Tableau Public
6	Performance	Fast dashboard loading with optimized data processing	Tableau Hyper Extracts (.hyper), Flask
7	Maintainability	Easy code maintenance through modular project structure	Python Flask, GitHub
8	Compatibility	Supports desktop and laptop browsers across platforms	Chrome, Edge, Firefox

4. PROJECT DESIGN

4.1Problem Solution Fit :

The **Problem–Solution Fit Canvas** helps validate whether the proposed solution effectively addresses the challenges faced by stakeholders in the food ordering industry. It identifies the target users, their business problems, existing approaches, and how the developed Flask-based Tableau Dashboard provides a practical and data-driven solution. The canvas also highlights customer behaviour, operational constraints, and the value delivered by the interactive analytics platform.

Block	Details
1. Customer Segment	Restaurant Owners, Business Analysts, Operations Managers, Marketing Teams, and Food Delivery Service Providers who need interactive business intelligence for analysing food ordering behaviour and customer trends.
2. The Problem	Difficulty in analysing large food ordering datasets, identifying customer preferences, monitoring business KPIs, and making informed operational decisions due to scattered and complex data.
3. Existing Alternatives	Manual analysis using Excel spreadsheets, static reports, traditional charts, and basic business reports with limited interactivity and filtering capabilities.
4. The Solution	A Flask-based web application integrated with Tableau Public dashboards that transforms raw food ordering data into interactive visualizations, KPI dashboards, and business intelligence reports.
5. Solution Features	Interactive Tableau dashboards, dynamic filters, KPI cards, customer behaviour analysis, city-wise and cuisine-wise comparisons, Tableau Story integration, and responsive web interface.
6. Customer Constraints	Limited technical expertise, time-consuming manual reporting, lack of centralized analytical tools, and the need for an easy-to-use web-based dashboard without programming knowledge.

Block	Details
7. Behaviour	Restaurant managers monitor revenue and orders, business analysts evaluate customer behaviour, marketing teams analyse demographics, and operations managers monitor delivery performance using business reports.
8. Triggers	Revenue decline, changing customer preferences, poor delivery performance, low customer ratings, seasonal demand fluctuations, and business review meetings.
9. Problem Root Cause	Large volumes of raw transactional data, absence of centralized business intelligence dashboards, manual reporting processes, and limited visualization capabilities.
10. Channels of Behaviour	**Online:** Flask Web Application, Tableau Public Dashboard, Business Portal. **Offline:** Business review meetings, restaurant operations planning, marketing strategy discussions, and management reporting.

4.2Proposed Solution :

The proposed solution is an interactive **Food Ordering Behaviour and Consumer Trends Dashboard** that combines the capabilities of **Python Flask** and **Tableau Public** to transform raw food ordering data into meaningful business intelligence. The system enables restaurant owners, business analysts, operations managers, and marketing teams to monitor key performance indicators (KPIs), analyse customer behaviour, evaluate delivery performance, and make informed business decisions through a responsive web application. By integrating interactive dashboards, dynamic filters, and data visualization techniques, the solution provides a centralized platform for analysing food ordering trends efficiently and accurately.

S.No.	Parameter	Description
1.	**Problem Statement**	Restaurant owners and business analysts find it difficult to analyse large food ordering datasets, monitor customer behaviour, identify business trends, and make informed decisions due to scattered data and the absence of interactive analytical tools.
2.	**Idea / Solution Description**	Develop a Flask-based web application integrated with Tableau Public dashboards that converts raw food ordering data into interactive visualizations, KPI dashboards, and business intelligence reports for effective decision-making.
3.	**Novelty / Uniqueness**	Combines Python Flask with embedded Tableau dashboards to provide a responsive analytics platform featuring dynamic filters, KPI cards, Tableau Story integration, and interactive business insights through a web interface.
4.	**Social Impact / Customer Satisfaction**	Improves restaurant operational efficiency, enhances customer satisfaction through better delivery performance, supports data-driven marketing strategies, and helps businesses understand customer preferences more effectively.

S.No.	Parameter	Description
5.	**Business Model (Revenue Model)**	Business Intelligence (BI) solution for restaurants and food delivery businesses. The platform can be offered as a subscription-based analytics service for restaurants, cloud kitchens, and food delivery organizations to support strategic decision-making.
6.	**Scalability of the Solution**	The modular Flask architecture and Tableau dashboard integration allow the application to support larger datasets, additional analytical modules, new KPIs, and future enhancements without major system redesign.

4.3Solution Architecture :

System Architecture & Data Flow

- **User Interface Layer:**
    - Users (Restaurant Owners, Business Analysts, and Managers) access the application through a **Web Browser**.
    - The **Flask Web Application** handles routing, user interactions, and dashboard integration.
    - Users apply filters such as **City, Cuisine, Restaurant Type, Customer Age, Ratings, and Order Value**.
- **Analytics & Visualization Layer:**
    - The system embeds the **Tableau Dashboard** using **tableau-2.min.js**.
    - Interactive dashboards, KPIs, charts, and **Tableau Story** provide business insights and trend analysis.

- **Data Processing & Storage Layer:**
  - The **Food Ordering Behaviour Dataset (CSV)** is the primary data source.
  - **Python (Pandas)** is used for data cleaning and preprocessing.
  - The processed data is visualized in Tableau, while project files are maintained in **GitHub**.

Architecture Goals

The solution architecture is designed to:

- Provide an interactive business intelligence platform for analysing food ordering behaviour.
- Deliver accurate KPIs and visual insights for better decision-making.
- Integrate Flask with Tableau for a responsive web-based dashboard.
- Support future enhancements and larger datasets through a scalable architecture.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning :

The project was developed using the **Agile Scrum methodology**, where the complete work was divided into multiple sprints. Each sprint focuses on implementing specific features such as user interface development, Flask integration, Tableau dashboard embedding, data processing, and interactive analytics. Story points were assigned based on the complexity and effort required for each task.

Product Backlog and Sprint Schedule :

Sprint	Epic	Story No.	User Story / Task	Story Points	Priority
Sprint-1	UI Setup	USN-1	Develop **base.html** framework with responsive Cream/Beige theme.	2	High
Sprint-1	Backend	USN-2	Configure Flask routing for home page and dashboard integration.	3	High
Sprint-2	Dashboard	USN-3	Embed Tableau dashboard using **tableau-2.min.js**.	5	High
Sprint-2	Data Processing	USN-4	Clean and prepare the food ordering dataset for visualization.	5	Medium
Sprint-3	Analytics	USN-5	Implement interactive filters, KPI cards, and dashboard analytics.	5	High

Sprint Progress Summary

The sprint progress summarizes the planned schedule, duration, and completion status of each sprint throughout the project development lifecycle.

Sprint	Total Points	Duration	Start Date	End Date (Planned)	Completed	Actual Release
Sprint-1	20	6 Days	01 July 2026	06 July 2026	20	06 July 2026
Sprint-2	20	6 Days	08 July 2026	13 July 2026	20	13 July 2026
Sprint-3	20	6 Days	15 July 2026	20 July 2026	—	—
Sprint-4	20	6 Days	22 July 2026	27 July 2026	—	—

Velocity Calculation

Velocity measures the amount of work completed during each sprint and helps estimate future sprint capacity. Based on the project schedule, the team's average velocity is calculated as follows:

$$\text{Average Velocity (AV)} = \frac{\text{Total Story Points}}{\text{Sprint Duration}}$$

$$AV = \frac{20}{6} \approx 3.33 \text{ Story Points per Day}$$

This indicates that the development team completed approximately **3.33 story points per day**, supporting effective sprint planning and predictable project delivery.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1Performance Testing :

Project Performance Testing Summary

- **Data Rendering & Preprocessing**:

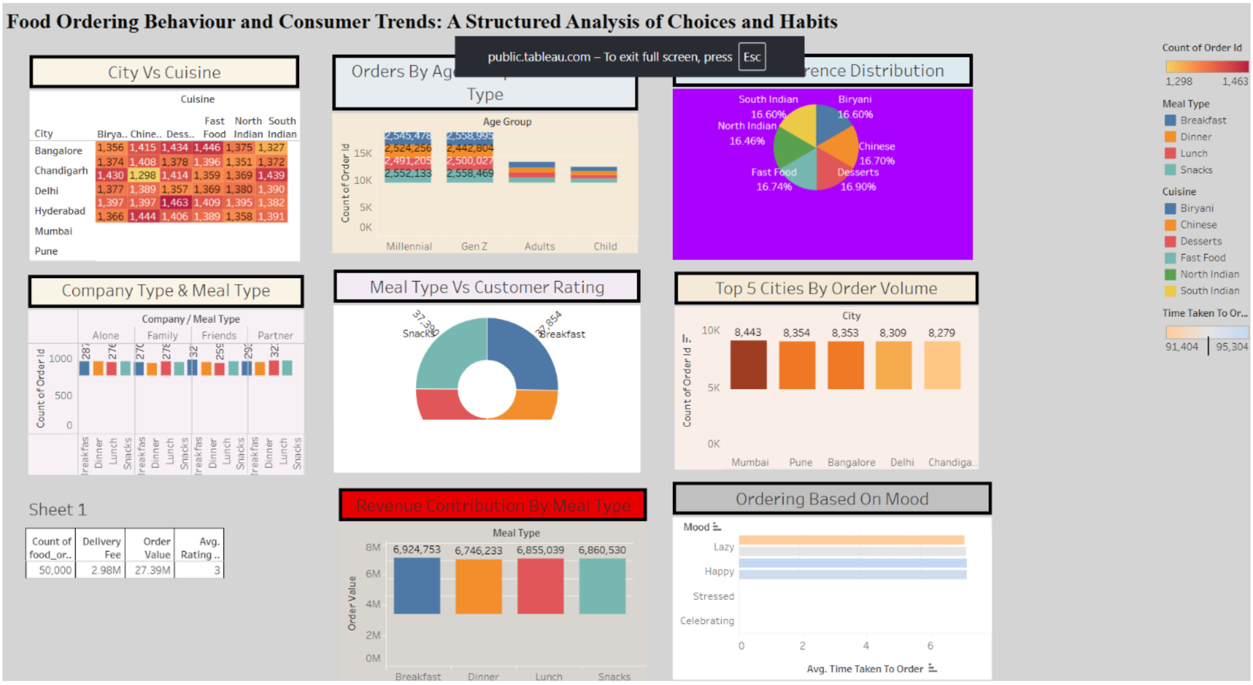
- The project involved uploading and rendering datasets in Tableau Public.
- Data preprocessing included removing null values, verifying data types, creating calculated fields, grouping Zipcodes (1–9), and validating renovation and condition fields.

- **Filters and Calculations**:
  - Filters were utilized with aggregation functions, primarily SUM(Usage) for total load mappings and AVG(Usage) to monitor baseline shifts.
  - Worksheets were designed with coordinated dynamic action filters for synchronized cross-filtering.
  - Calculated fields were implemented to solve Tableau's order of operations conflict, allowing for dynamic ranking of states within specific geographical region filters.
- **Visualization and Story Design**:
  - The "Dashboard 1" layout includes metrics such as Total Orders (50,000), Revenue (₹27,386,555.00), and Average Order Value (₹547.73).
  - Visualizations include distributions for delivery time, cuisine, meals, and customer age, alongside revenue by city and ratings.
  - A Tableau Story titled "FOOD ORDERING BEHAVIOUR" was created, containing multiple story points focusing on cities, revenue, cuisine, order values, meals, and age distribution.

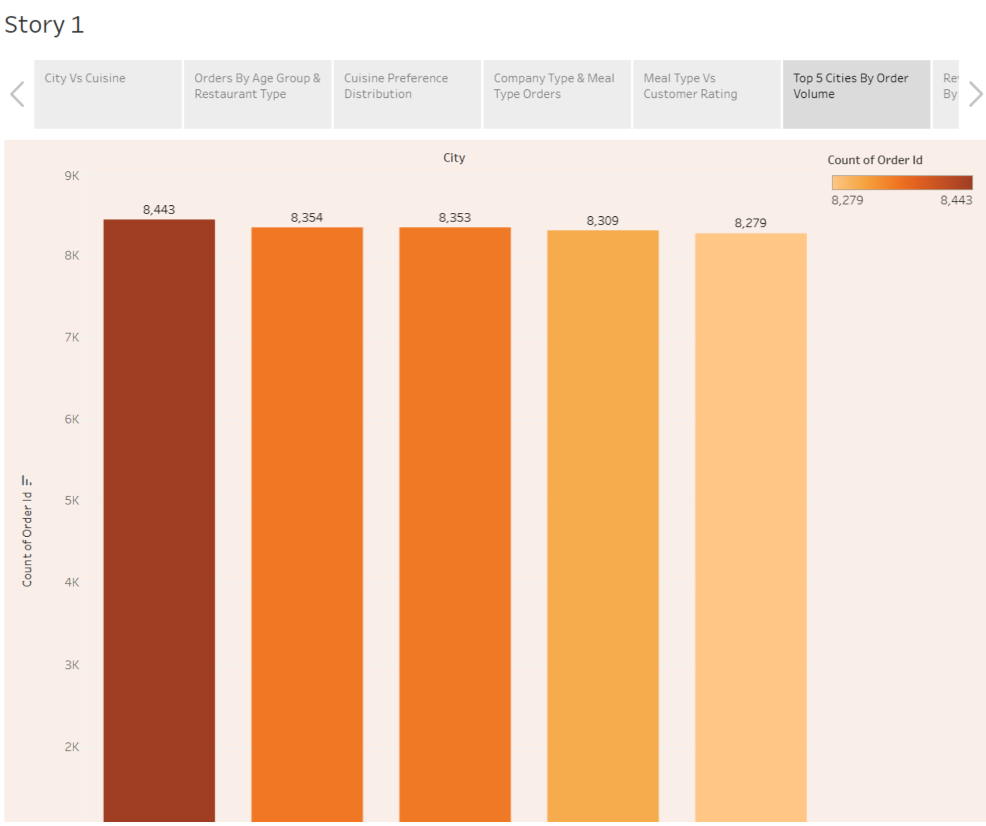
7. RESULTS

7.1 Output Screenshots :

Dashboard Screenshot :



Story Screenshot :



8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides a centralized, interactive dashboard for analysing food ordering behaviour and customer trends.
- Helps restaurant owners identify high-performing cities, cuisines, and restaurant categories quickly.
- Enables business analysts to monitor KPIs such as orders, revenue, ratings, and delivery performance.
- Supports marketing teams in analysing customer demographics and purchasing behaviour for targeted campaigns.
- Integrates Flask with Tableau to provide a user-friendly web-based analytics platform.

Disadvantages

- The accuracy of insights depends on the quality and completeness of the input dataset.
- Tableau Public stores dashboards publicly, making it unsuitable for confidential business data.
- The current system analyses historical data and does not provide real-time order tracking.
- Advanced predictive analytics and machine learning features are outside the scope of this project.
- The application currently requires internet access to load the embedded Tableau Public dashboard.

9. CONCLUSION

This project transformed the Food Ordering Behaviour dataset into an interactive business intelligence dashboard using Flask and Tableau Public. The dashboard provides clear insights into customer behaviour, revenue, delivery performance, ratings, and cuisine trends through interactive visualizations and KPI cards. It enables restaurant owners and business analysts to make data-driven decisions, improve operational efficiency, and identify business opportunities quickly through an easy-to-use web application.

10. FUTURE SCOPE

Future Scope

- Integrate real-time food ordering data using APIs for live dashboard updates.
- Add machine learning models to predict customer behaviour, delivery time, and order trends.
- Include customer review sentiment analysis for deeper business insights.
- Expand the dashboard to support additional cities, restaurants, and larger datasets.
- Implement secure user authentication with role-based access for restaurant owners, analysts, and managers.

11. APPENDIX

Source Code(if any) :

All source files are maintained in the GitHub repository listed below .

Dataset Link :

https://github.com/ssaisiddardha/Food-Ordering-Behaviour-and-Consumer-Trends/blob/main/Templates/5.%20Project%20Development%20Phase/food_ordering_behavior_dataset.csv

GitHub & Project Demo Link :

GitHub Repository: https://github.com/ssaisiddardha/Food-Ordering-Behaviour-and-Consumer-Trends

Tableau Public Profile :

LINK :
https://public.tableau.com/app/profile/sai.siddardha.somarouthu/vizzes

Team Members : Gubbala Durga Vinay
Eswara Sai Dumpala
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- THANK YOU -